Module 6: Searching Techniques

6.1 Linear Search, Binary Search, Hashing-Concept, Hash functions.

6.2 Collision Resolution Techniques

* Linear Search

It is one of the conventional searching techniques that sequentially searches for an element in the list. It typically starts with the first element in the list and moves towards the end in a step-by step fashion. In each iteration, it compares the element to be searched with the list element and if there is a match, it returns the location of the list element

Algorithm: Linear Search (A, N, Val)

```
Step 1: Set pos = -1
Step 2: Set i = 1
Step 3: Repeat step 4 while (i <= n)
Step 4: If A[i] = val
            set pos = i
            print pos
            "Element found" and go to step 6
        else
            set i = i+1
        [end of if]
    [end of loop]
Step 5: If pos = -1,
            print " Element not found".
Step 6: Exit
```

i
↓

arrayA

A	B	C	D	E
1	2	3	4	5

A → array
H → max element → 5
Val → D

Advantages.

1. It is a simple searching technique that is easy to implement.
2. It does not require the list to be sorted in a particular order.

Disadvantages

1. It is efficient for large data. The time complexity is O(n) and for large data, time increases.

* Binary search.

Binary search is another kind of searching techniques which requires the array to be sorted in an ascending order. It Follows the divide and break rule.

1) It searches the middle element
2) If value < middle element
   it searches the lowerbound (elements in the array before middle element).
3) If value > middle element
   it searches the upperbound (elements after middle element)

Algorithm: Binary search (A, lowerbound, upperbound, val)

Step 1: Set beg = lowerbound
        Set end = upperbound
        Set pos = -1

Step 2: Repeat steps 3 & 4 while beg <= end.

Step 3: set mid = (beg + end) / 2

Step 4: if A[mid] = val
            Set pos = mid
            print pos
            print "Element found"
        else if A[mid] > val
            set end = mid-1
        else
            set beg = mid + 1
        [end of if]. [end of loop]

Step 5: if pos = -1
            print "element not found"
        [end of if]

Step 6: exit.

Advantages

1. requires less number of iterations
2. Lot faster than linear search.

Disadvantages

1. requires array to be sorted.
2. little difficult to implement.

Time complexity $O(\log_2 n)$

Example

10	20	30	40	50
0	1	2	3	4

① val = 30

beg = 0
end = 4
mid = 2
Element found at 2
A [2] = 30

② val = 50

① beg = 0
end = 4
mid = 2
A[2] = 30 < 50

② beg = 3 end = 4 mid = 3.
A[3] = 40 < 50.

③ beg = 4 end 4 mid = 4
A[4] = 50
Element found

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③ val = 45.

(Iterations from ②).

Element not found

* Hashing.

Hashing finds the location of an element in a data structure without making any comparisons. Hashing uses a mathematical function to determine the location of an element called as hash function.

Different hash functions.

1. Division method
2. Multiplication method
3. Mid Square method
4. Folding method.

1 Division method

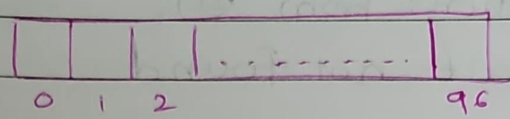
$h(x) = x \bmod M.$

$x \rightarrow$ key value

$M \rightarrow$ size of hash table ie array size

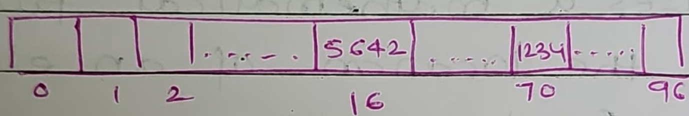
Q1. Consider the hash values of keys 1234 and 5642 Consider $M = 97$.

Ans



$x = 1234$, $h(x) = 1234 \bmod 97$
$= 70$

$x = 5642$, $h(x) = 5642 \bmod 97$
$= 1.6$



2 Multiplication method.

The steps involved in multiplication method are as follows.

1. Choose a constant A such that $0 < A < 1$
2. Multiply the key k by A.
3. Extract the fractional part of kA
4. Multiply the result of step 3 by size of hash table n.

$h(k) = \lfloor m(kA \bmod 1]$

$A \rightarrow 0.618033 = (\text{Sqrt } 5 - 1)/2$

Q2 Given a hash table of size 1000 map the key 12345 to a appropriate location in the hashtable

Ans. $M = 1000$

$KA = 12345 \times 0.618033$

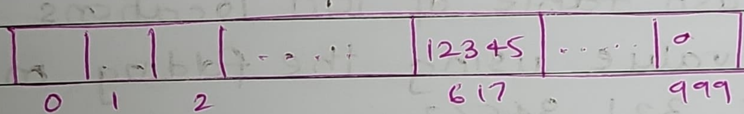
$= 7629.617385$

$hk = \lfloor 1000\ (KA \bmod 1) \rfloor$

$= \lfloor 1000 \times \cancel{7629}.617385 \rfloor$

$= 617.385$

$= 617$



Q3. Mid square method.

1. square the value of key K^2.
2. Extract the middle r digits of the result obtained from step 1.

Q3. Calculate the hash value for keys 1234 and 5642 using the mid square method and the hash table has 100 memory locations.

Ans $M = 100$

$h(K) = h(1234)$

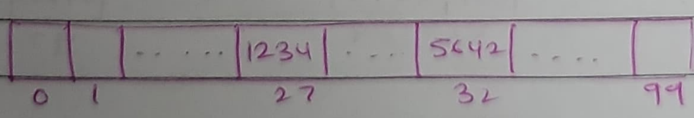
$(1234)^2 = 1522756$

Middle digit $= 27$

$h(K) = h(5642)$

$5642^2 = 31832164$

Middle digit $= 32$



4. Folding method

1. Divide the key value into number of parts ie $k_1, k_2, \ldots, k_n$ where each part has the same number of digits except the last part which may have lesser digits than the other parts.
2. Add the individual parts ie $k_1 + k_2 + \ldots + k_n$. The Hash value is produced by ignoring the last carry if any.

Q4. Given a hash table of 100 locations. Calculate the hash value using the folding method for key 5678, 321, 34567.

Ans ① h(5678) = 56 78

Addition = 134

Hash value = 34

② h(321) = 32 1

Addition = 33

Hash value = 33

③ h(34567) = 34 56 7

Addition 97

Hash value = 97